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SUMMARY

The spatio-temporal properties of seismicity give us incisive insight into the stress state evolution and fault structures of the crust. Empirical models based on self-exciting point processes continue to provide an important tool for analysing seismicity, given the epistemic uncertainty associated with physical models. In particular, the epidemic-type aftershock sequence (ETAS) model acts as a reference model for studying seismicity catalogues. The traditional ETAS model uses simple parametric definitions for the background rate of triggering-independent seismicity. This reduces the effectiveness of the basic ETAS model in modelling the temporally complex seismicity patterns seen in seismic swarms that are dominated by aseismic tectonic processes such as fluid injection rather than aftershock triggering. In order to robustly capture time-varying seismicity rates, we introduce a deep Gaussian process (GP) formulation for the background rate as an extension to ETAS. GPs are a robust non-parametric model for function spaces with covariance structure. By conditioning the length-scale structure of a GP with another GP, we have a deep-GP: a probabilistic, hierarchical model that automatically tunes its structure to match data constraints. We show how the deep-GP-ETAS model can be efficiently sampled by making use of a Metropolis-within-Gibbs scheme, taking advantage of the branching process formulation of ETAS and a stochastic partial differential equation (SPDE) approximation for Matérn GPs. We illustrate our method using synthetic examples, and show that the deep-GP-ETAS model successfully captures multiscale temporal behaviour in the background forcing rate of seismicity. We then apply the results to two real-data catalogues: the Ridgecrest, CA 2019 July 5 M_w 7.1 event catalogue, showing that deep-GP-ETAS can successfully characterize a classical aftershock sequence; and the 2016–2019 Cahuilla, CA earthquake swarm, which shows two distinct phases of aseismic forcing concordant with a fluid injection-driven initial sequence, arrest of the fluid along a physical barrier and release following the largest M_w 4.4 event of the sequence.

Key words: Inverse theory; Statistical methods; Statistical seismology.

1 INTRODUCTION

The study of seismicity patterns is a core concern of observational seismology. By documenting seismicity in the form of catalogues, we aim to better understand the mechanics of the crust as it evolves through time. Increasing progress in observational seismological instrumentation and methodology have led to increasingly complete and accurate catalogues, which have revealed fascinating fine-scale structure in seismicity (Wei *et al.* 2011; Shelly *et al.* 2016; Ross *et al.* 2019, 2020, e.g.). However, as a comprehensive physical theory of seismicity remains elusive, we still rely heavily on empirical statistical models to understand catalogues of seismic events. The de facto standard statistical model for the analysis of seismicity at short timescales is the epidemic-type aftershock sequence (ETAS) model, introduced by Ogata (1988) for temporal catalogues, and extended to spatio-temporal catalogues by Ogata (1998). The ETAS model explicitly combines a background rate of earthquake activity, independent of past events, with self-excitation of aftershocks following the combined Omori–Utsu (Utsu 1961) aftershock decay law and an earthquake productivity law developed by Ogata (1998) in reference to the earthquake catalogue classification schema of Utsu (1971). The ETAS model allows us to decouple the background forcing of seismicity from self-excitation (i.e. it declusters earthquake catalogues), and allows for statistical forecasting of earthquake rates into the future once calibrated.

The spatio-temporal variations in the ETAS parameters have given useful insight into the evolution of seismicity. Whilst classical aftershock sequences are normally well modelled by the self-triggering component of ETAS, swarm-type sequences are insufficiently

clustered to conform to ETAS with a stationary forcing rate, and instead reflect aseismic changes in forcing (Kumazawa & Ogata 2014, e.g.). In the zero-clustering limit, the behaviour of swarms may be modelled as an inhomogenous Poisson process; however, in reality some degree of clustering must be accounted for, which poses a difficult challenge as it is impossible to directly observe both the self-excitation rate and aseismic forcing rate independently. Many methodologies have been proposed to overcome this challenge. For instance, Marsan *et al.* (2013) proposed deriving a time-dependent aseismic forcing rate by iteratively obtaining the maximum-likelihood estimate (MLE) parameters of an ETAS model with per-earthquake forcing rate, updating the forcing given the MLE parameters, and then temporally smoothing the resulting rate to avoid overfitting—this method was applied to the analysis of the foreshock sequence of the M_w 9.0 2011 Tohoku earthquake in Marsan & Enescu (2012). Hainzl *et al.* (2013) then used the method proposed in Marsan *et al.* (2013) to investigate the impact of transients in the aseismic forcing rate on the prediction of aftershock productivity, an important consideration for accurate short-term forecasting of seismic activity. Kumazawa *et al.* (2016) investigated the prolific Izu Peninsula swarms, and found a close fit between gradient-penalized piecewise-linear functions and a simple model based on exponential decay of strains recorded at Higashi-Izu which allowed them to infer synchronization between earthquake swarm activity and volumetric strain changes induced by magma intrusions. These successes in analysis motivate us to investigate improvements in methodology that can robustly quantify the uncertainty in estimates of aseismic forcing rates.

In general, as we have no knowledge of the functional form of background variability, it is preferable to use a non-parametric formulation for aseismic forcing; however, solving for a non-parametric background rate model has traditionally been difficult and does not permit a Hessian based estimate of uncertainty at the MLE. Recent studies (Ross 2021; Molkenhuth *et al.* 2022) have advocated for a fully Bayesian approach, in which the background rate and ETAS parameter priors are specified, combined with the ETAS likelihood to produce a posterior probability distribution, and sampled using Markov Chain Monte Carlo methods (MCMC). While MCMC is in general an expensive technique, it is an unfortunate necessity for the ETAS problem due to the lack of a general closed-form conjugate model that would allow direct draws from the posterior distribution. Molkenhuth *et al.* (2022), in particular, show how the background rate may be successfully modelled in non-parametric fashion by making use of Gaussian processes (GP) to specify the background prior and using a branching-structure formulation of the ETAS model to decouple the background events from aftershocks, which are handled by self-excitation. Beyond the confines of the ETAS model, GPs have also proven useful in the spatial analysis of seismic sequences. Bayliss *et al.* (2020) investigated numerous geophysical covariates for seismicity in a log-Gaussian Cox process model of spacial seismicity (e.g. distance to mapped faults, strain rates, etc.), and used a GP to handle additional unspecified seismic intensity rate variability. Ross *et al.* (2022) investigated the hyperparameters of fully GP specified log-Gaussian Cox process models for seismic sequences as a quantitative measure of geometrical properties such as the anisotropy of seismicity. GPs are a flexible and robust framework for non-parametric modelling, however in the context of geophysical inversion problems they suffer from two major deficiencies. The first is that, without approximations, they scale poorly to large problem sizes (e.g. large seismic catalogues), as specifying the covariance structure requires inverting a matrix at $O(n^3)$ expense where n is the catalogue size. The second is that basic GP analysis assumes homogeneity of the underlying statistical distribution of the prior—this assumption poorly models abrupt spatio-temporal changes in behaviour that are found in the real Earth.

In this study, our contributions are as follows. We show how the aforementioned issues may be resolved by making use of a deep GP model (DGP) approximated by the solution to a stochastic partial differential equation (SPDE, Lindgren *et al.* (2011)). We also improve the rate of MCMC convergence by making use of appropriate modern samplers that exploit the structure of the SPDE approximation and the gradient of the combined semi-parametric model posterior. Finally, we illustrate the DGP-ETAS model using a combination of synthetic catalogues, the Ridgecrest, CA earthquake sequence, and the Cahuilla, CA earthquake swarm.

2 SAMPLING THE ETAS MODEL

The temporal ETAS model is a self-exciting marked point process defined on the space $\mathcal{T} \times \mathcal{M}$, where $\mathcal{T} = (0, T)$ is the time domain of interest and $\mathcal{M}$ is the mark space of the process, which for ETAS is the event magnitude. Realizations of the process are earthquake catalogues $\mathcal{C}$ where the i th event is defined by the tuple (t_i, m_i) . $\mathcal{C}$ is endowed with a cut-off magnitude m_0 , which is the lower limit of observability. The temporal ETAS model may be defined by its conditional intensity function, which gives the instantaneous rate of earthquakes given the background forcing and past events. Assuming that earthquake triggering is time-independent, the conditional intensity is the sum of the background rate and the triggering:

$$\lambda(t|\mathcal{C}, \mu, \theta) = \mu(t) + \sum_{i \in \mathcal{C} | t_i < t} \phi(t - t_i | m_i, \theta). \quad (1)$$

The most common parametrization of ϕ , and the one we use in this work, is

$$\phi(t - t_i | m_i, \theta) = \kappa(m_i | \theta) g(t - t_i | \theta) \quad (2)$$

$$\kappa(m_i | \theta) = K \exp(\alpha(m_i - m_0)) \quad (3)$$

$$g(t | \theta) = (t + c)^{-p}. \quad (4)$$

With this definition, the ETAS model triggering parameters are $\theta = (K, \alpha, c, p)$. The magnitude term κ is proportional to the Utsu aftershock-productivity law, while the temporal term g is proportional to the modified Omori–Utsu aftershock decay law. The log-likelihood of observing

catalogue $\mathcal{C}$ is given by

$$\log p(\mathcal{C}|\mu, \theta) = \sum_{i \in \mathcal{C}} \log(\lambda(t_i|\mathcal{C}, \mu, \theta)) - \int_0^T \lambda(t|\mathcal{C}, \mu, \theta) dt. \quad (5)$$

Given that the conditional-likelihood itself contains a sum over the catalogue, the log-likelihood of the standard ETAS model thus has a double loop, and the effort required to compute it scales with the number of events squared. When considering heterogeneous background rates, the problem is even worse as the background parameters influence the ETAS likelihood, dramatically increasing the complexity and dimensionality of the likelihood surface. Following Ross (2021), a useful reformulation that improves computational tractability is to make use of the splitting property of Poisson processes (Veen & Schoenberg 2008). This formulation of ETAS introduces the auxiliary branching variables $\{B_i\}_{i=1}^N$, where $B_i = j$ tells us that the i th event in $\mathcal{C}$ had ‘parent’ j and $B_i = 0$ tells us that the i th earthquake was a ‘background’ event. This factorization splits the estimation of the background rate and the ETAS parameters into two subproblems, where the background catalogue is described by an inhomogeneous Poisson process and the aftershocks are handled by ETAS without background forcing—this greatly simplifies inference of all parameters, reduces the expense of computing the ETAS likelihood and serendipitously declusters the catalogue as a byproduct of sampling. We introduce subcatalogues $\mathcal{C}_j = \{i|B_i = j\}$ so that $|\mathcal{C}_j|$ is the number of earthquakes triggered by event j (or the background for $\mathcal{C}_0$). The conditional intensity for the i th event then becomes

$$\lambda(t_i|\mathcal{C}, \mu, \theta, B_i) = \begin{cases} \mu(t_i) & B_i = 0 \\ \phi(t_i - t_{B_i} | m_{B_i}, \theta) & 1 \leq B_i < i, \end{cases} \quad (6)$$

and the log-likelihood function becomes

$$\begin{aligned} \log p(\mathcal{C}|\mu, \theta, B) = & \sum_{i \in \mathcal{C}_0} \log(\mu(t_i)) - \int_0^T \mu(t) dt \\ & + \sum_{j \in \mathcal{C}} \left[|\mathcal{C}_j| \log \kappa(m_j|K, \alpha) - \kappa(m_j|K, \alpha) G(T - t_j|c, p) + \sum_{i \in \mathcal{C}_j} \log g(t_i - t_j|c, p) \right], \end{aligned} \quad (7)$$

where

$$G(T - t_j|c, p) = \int_0^{T-t_j} g(t|c, p) dt. \quad (8)$$

Note that the double sum in this form does not suffer quadratic scaling because each earthquake has a unique parent and so is only ever visited once in the interior sum. Marginalizing over the branching variables gives us the probability distribution of μ and θ conditioned on all possible combinations of parent/child earthquake pairs (i.e. with the likelihood defined by eq. 5). The conditional probability for the branching variables is analytic, and is given by

$$p(B_i = j|\mu, \theta) = \begin{cases} \frac{\mu(t_i)}{Z_i} & j = 0 \\ \frac{\kappa(m_j|K, \alpha) g(t_i - t_j|c, p)}{Z_i} & 1 \leq j < i, \end{cases} \quad (9)$$

where the normalization factor is

$$Z_i = \mu(t_i) + \sum_{j < i} \kappa(m_j|K, \alpha) g(t_i - t_j|c, p). \quad (10)$$

3 DEEP GAUSSIAN PROCESSES

With the general form of the latent branching-variable ETAS model specified, it comes time to specify the model for the time-variable background rate $\mu(t)$. Given that the ETAS model is a general phenomenological description of seismicity, it is appropriate to model the background rate with a general function. We use the framework of GPs as a robust non-parametric description of the background seismicity rate (Rasmussen & Williams 2006; Molkenhain *et al.* 2022). GPs act as priors over function space (in our case the space of continuous functions on the interval $[0, T]$), and are defined by the property that the distribution $\{f(x_i)\}$ of a GP f evaluated at any finite collection of points $\{x_i\}$ is given by a multivariate normal with mean function $m(x)$ and covariance function $C(x, x')$, that is,

$$\begin{bmatrix} f(x_1) \\ f(x_2) \\ \vdots \\ f(x_n) \end{bmatrix} \sim N \left(\begin{bmatrix} m(x_1) \\ m(x_2) \\ \vdots \\ m(x_n) \end{bmatrix}, \begin{bmatrix} C(x_1, x_1) & C(x_1, x_2) & \dots & C(x_1, x_n) \\ C(x_2, x_1) & C(x_2, x_2) & \dots & C(x_2, x_n) \\ \vdots & \vdots & \ddots & \vdots \\ C(x_n, x_1) & C(x_n, x_2) & \dots & C(x_n, x_n) \end{bmatrix} \right). \quad (11)$$

The mean and covariance functions completely specify the GP. The covariance function, which specifies the regularity and characteristic length-scales of the GP, is the central object in GP modelling. A prototypical example would be the squared exponential covariance

$$C(x, x') = \sigma \exp \left[-\frac{\|x - x'\|^2}{2l^2} \right], \quad (12)$$

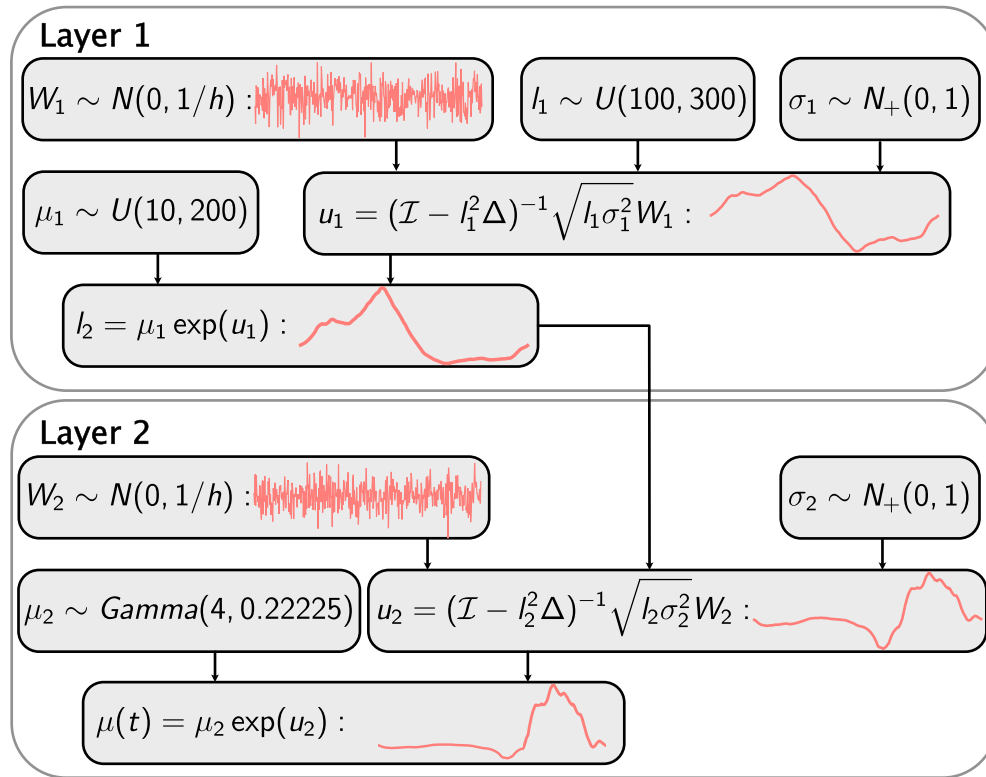


Figure 1. Schematic of a two-layer DGP based on the covariance operator definition. Further layers may be similarly concatenated to produce deeper networks, however the greater expressive capacity is unlikely to be required in an inverse problem context. Parameters without a representative trace are scalars, those with traces are functions defined on the interval of interest. The parameter values are those used for the synthetic square catalogue described in the examples section. The schematic is arranged sequentially downward, so that each line feeds into the one below.

which generates functions f that are infinitely differentiable and have characteristic length-scale l and amplitude σ . Another commonly used function is the Matérn covariance, which is ubiquitous in geospatial statistics. The Matérn has the form

$$C(x, x') = \sigma^2 \frac{2^{1-\nu}}{\Gamma(\nu)} \left(\sqrt{2\nu} \frac{\|x - x'\|}{l} \right)^\nu K_\nu \left(\sqrt{2\nu} \frac{\|x - x'\|}{l} \right), \quad (13)$$

where the Γ is the generalized factorial or Gamma function, K_ν is the modified Bessel function of the second kind and ν is the regularity parameter. Functions that have Matérn covariance are differentiable $[\nu] - 1/2$ times. In particular, the squared exponential covariance is the special case of the limit of the Matérn covariance as ν goes to infinity. The added roughness of lower ν Matérn covariance typically better represents realistic phenomena. For data with Gaussian noise, GP regression models have a closed form solution once the mean and covariance are fully specified, and careful selection of the form of the covariance matrix can lead to rich and fully interpretable models with a high degree of predictive power (Rasmussen & Williams 2006). However, while GP regression models have analytic solutions, inverse problems do not in general, as is the case with the estimation of the aseismic forcing rate $\mu(t)$ considered here. Furthermore, the construction of suitable covariance kernels in a classical-GP framework requires a substantial degree of domain expertise, and consequently a large fraction of the GP modelling literature assumes covariance functions that are homogeneous and isotropic for simplicity (see e.g. Gelfand & Schliep 2016 for a brief review). The loss of model flexibility inherent in this assumption can result in substantial misfit and prediction error, either due to underfitting or overfitting. Because of these limitations, learning methods for automatic discovery of appropriate covariance functions have become a major focus of GP research in order to improve computational efficiency and reduce domain expertise requirements, whilst maintaining the attractive features of GP robustness and interpretability.

In this work, we investigate DGPs as a means of learning appropriate covariance functions. In the DGP framework, multiple GPs are chained together, either feeding the outputs of one GP to the inputs of another (Damianou & Lawrence 2013), or alternatively by letting the outputs of one GP change the covariance function of the next (Roininen *et al.* 2016, see Fig. 1). The latter approach is particularly appealing, because the covariance function can be easily evaluated for each layer of the DGP, resulting in a fully interpretable model. Additionally, by making use of an SPDE approximation to GPs, DGP models can be very efficiently simulated, as we will describe below.

3.1 Approximating DGPs with stochastic partial differential equations

GP models are expressive, robust, easily interpretable and easily implementable—however, the use of pure GP models in large-scale inverse problem settings has been limited. This is due to their inherently poor scaling. The evaluation of a GP model requires inversion of an $n \times$

n covariance matrix for a GP conditioned on n points. The construction of this matrix scales in memory like $O(n^2)$ and the computational complexity of inversion scales like $O(n^3)$. As a result, GP approximation methods are an active area of research. The key work of Lindgren *et al.* (2011) established a link between GP models using the Matérn class of covariance functions and the solution to an SPDE. The advantage of this approach is that the linear system required to solve the SPDE is extremely sparse, becoming tridiagonal in the case of an approximate 1-D Matérn. As a result, sampling from the approximate GP defined by the SPDE solution is vastly cheaper than the full GP. Both large-scale problems and nonlinear inverse problems (requiring many GP evaluations) become feasible, when the full GP solution would not be.

To draw samples for the 1-D inhomogeneous Matérn-1/2 process using the SPDE approximation, we solve the following for u :

$$(I - l^2 \Delta)u = \sqrt{l\sigma^2}W, \quad (14)$$

where W is a white noise process and Δ is the Laplacian operator. The SPDE is solved by approximating it by the finite-difference method, which gives rise to a tridiagonal Laplacian. The boundary conditions (usually Neumann) bias the solution to the SPDE near the edges, so typically the domain is extended so that the boundaries are not near the time interval of interest. Explicitly, the system that is solved for u at each layer of the DGP is

$$\begin{bmatrix} 1 + 2\frac{l_1^2}{h^2} & -\frac{2}{h^2} & \dots & 0 & 0 \\ -\frac{1}{h^2} & 1 + 2\frac{l_2^2}{h^2} & \dots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \dots & 1 + 2\frac{l_{n-1}^2}{h^2} & -\frac{1}{h^2} \\ 0 & 0 & \dots & -\frac{2}{h^2} & 1 + 2\frac{l_n^2}{h^2} \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_{n-1} \\ u_n \end{bmatrix} = \begin{bmatrix} \sqrt{l_1\sigma^2} & 0 & \dots & 0 & 0 \\ 0 & \sqrt{l_2\sigma^2} & \dots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \dots & \sqrt{l_{n-1}\sigma^2} & 0 \\ 0 & 0 & \dots & 0 & \sqrt{l_n\sigma^2} \end{bmatrix} \begin{bmatrix} W_1 \\ W_2 \\ \vdots \\ W_{n-1} \\ W_n \end{bmatrix} \quad (15)$$

For a given fixed-length discretization with grid spacing h , the white noise process W on the grid can be generated by sampling a vector of unit normals w and scaling it by $W = w/h$, where the $1/h$ factor ensures that the average integrated power of the process is 1 per unit length, which in turn means the characteristic amplitude of the SPDE solution is σ . We designate the solution, given l, σ , and a particular realization of white noise forcing w as $u = S[w, l, \sigma]$. For DGP models, the length-scale term l is itself a function of time after the first layer: $l = l(t)$. To approximate the multilayer DGP model using an SPDE approach, we set the length-scale for the first layer l_1 to be uniform (where the index now refers to the DGP layer being considered), and then generate $u_1 = S[w_1, l_1, \sigma_1]$ by solving the SPDE with a sample of white noise forcing w_1 . We then set the length-scale for the second layer $l_2 = \mu_1 \exp(u_1)$ for a scale constant μ_1 , making l_2 strictly positive, and obtain $u_2 = S[w_2, l_2, \sigma_2]$ using white noise w_2 , etc. The background rate of seismicity is also strictly positive. For the DGP-ETAS model, the background rate $\mu(t)$ is therefore given by $\mu(t) = \mu_1 \exp(u_1)$ for a one layer model, $\mu(t) = \mu_2 \exp(u_2)$ for a two-layer model and so on. For an n layer model, the scale constants μ_i have units of length for $i < n$ as they correspond to an average lengthscale of layer $i + 1$, and units of inverse time for $i = n$ as they correspond to the average background rate. This iteration can be carried out to arbitrarily high order, however, as is carefully investigated in Dunlop *et al.* (2018), the additional expressivity of non-uniform lengthscales saturates for more than around three layers. For inverse problems where the posterior uncertainty is likely to be high, such as the DGP-ETAS model investigated here, a two-layer DGP provides the best trade-off between computational complexity and expressivity. Fig. 1 shows the flow of information in an example two-layer DGP.

4 MODEL PRIORS

Parameter prior specification is a crucial part of Bayesian modelling. Given data y and parameters θ , Bayes' rule tells us that the posterior is proportional to the likelihood of observing the data given the parameters, multiplied by the prior distribution of those parameters: $p(\theta|y) \propto p(y|\theta)p(\theta)$. When the data are very informative (i.e. $p(y|\theta)$ is sharply peaked), then the prior typically has little impact on our final interpretation, however if the data are uninformative, the prior can substantially affect the results. It is therefore important that the prior is specified in a way that ensures the posterior is well behaved. For the DGP-ETAS model, we consider all of the model parameters to be uncorrelated, *a priori*, and discuss our choice of priors below.

4.1 ETAS parameters

The four ETAS parameters, K, α, c and p are all strictly positive (in particular, $p > 0$). α, c and p generally have 'typical' ranges for a particular setting, whereas K instead depends on the lower magnitude cutoff of the catalogue. We use Inverse-Gamma priors for α, c and the auxiliary variable $\tilde{p} = p - 1$, which we tune to place 98 percent of the probability mass within a designated typical range for the relevant problem. For K , we use a normal distribution truncated between 0 and an upper bound K_{upper} in order to suppress an effect in the MCMC sampling that traps chains in an infeasible part of model space—this occurs when K is large and all earthquakes after the first in the catalogue are ascribed to triggering rather than background effects. For the constant background rate reference ETAS model, we use a Gamma distribution prior for μ , as the conditional posterior for μ is then given in closed form by $p(\mu|K, \alpha, c, p, B, C) = \text{Gamma}(a + |C_0|, 1/(b + T))$ where a and b are the parameters of the Gamma prior; we always estimate $a = |C|/T$ and $b = |C|/2T$.

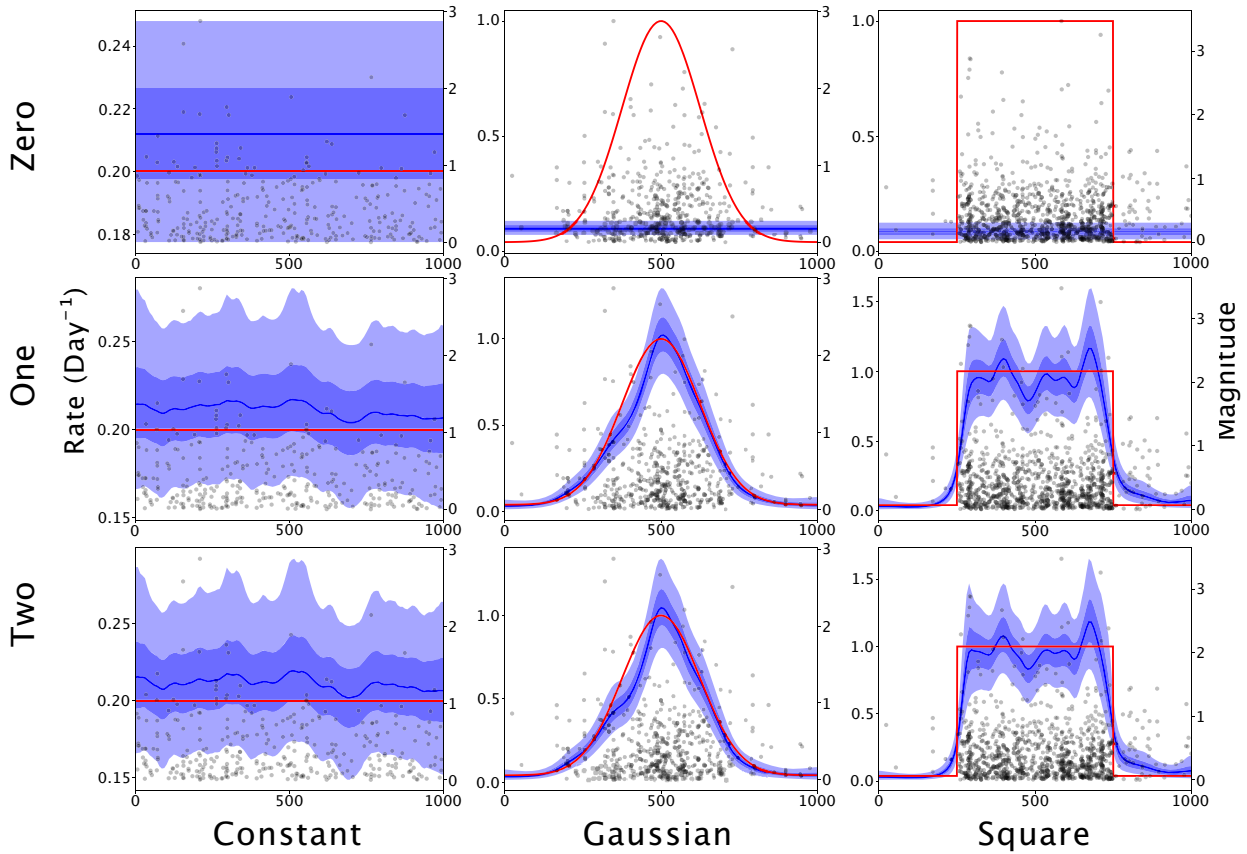


Figure 2. Results of inversions for the background rate of synthetic catalogues; the output matrix contains the combinations for zero, one and two layer DGP models, for constant, Gaussian and square perturbations of the background rate. Note that the zero-layer DGP corresponds to a normal ETAS model. Catalogue events are shown as grey circles, the ground truth background rate as the red line, and the 90 percent credible interval, 50 percent credible interval and median of the posterior are shown in increasingly dark shades of blue.

4.2 DGP parameters

The white noise parameters w_i are given a standard multivariate Normal prior with identity covariance $w_i \sim N(0, 1)$. Length-scale parameters (e.g. l_1 , and μ_1 for the two-layer model) have Uniform priors; for multilayer models we have found it is normally useful to have the ranges of the length-scales be disjoint so that the different GP layers capture variability at different scales—for instance $l_1 \sim U(100, 300)$ and $\mu_1 \sim U(10, 200)$ for the DGP-ETAS model in Fig. 2. The rate parameter (μ_1 for the one layer model and μ_2 for the two layer model) is given a Gamma prior for consistency with the basic ETAS model. The GP variance parameters σ_i are given standard half-normal priors $N_+(0, 1)$ (i.e. a zero-mean Gaussian with standard deviation 1, truncated to be strictly positive).

5 SAMPLING SCHEME

While the full DGP-ETAS model is quite complicated, the parameters of the model naturally partition into blocks that are conditionally simple to sample from, if the other blocks are held fixed. As such, we use a Metropolis-within-Gibbs scheme to draw samples from the posterior distribution for the DGP-ETAS model. In Metropolis-within-Gibbs, each block of parameters is updated sequentially (Gibbs sampling), with the update conditional on the other blocks. For each block update, a new set of parameter values is proposed using an appropriate sampling method for that block, and then the proposal is accepted or rejected according to a rule that guides results in sampling the posterior (a Metropolis update). The specific form of the rule depends on the way the samples are proposed.

To sample both the ETAS scalar parameters K , α , c , p and the GP hyperparameters l_i , μ_i and σ_i , we use Hamiltonian Monte Carlo (HMC, Neal 2011; Betancourt 2017). HMC utilizes gradient information to accelerate the traversal of the posterior, resulting in samples with low autocorrelation, and is a good general-purpose choice for MCMC sampling nonlinear problems. We use forward-mode automatic differentiation to obtain the gradient of the derivative in respect to the scalar parameters. For the white-noise vectors w_i , we use elliptical slice sampling (ESS; Murray *et al.* 2010), which is a particularly efficient way to sample the conditional posterior when the prior distribution is a multivariate normal, which is always the case for w_i . The branching variables B can be exactly sampled given the conditional distribution defined in eq. (9). For the two-layer DGP-ETAS model, we then have the following algorithm for drawing posterior samples, where for notational convenience we write Θ_ξ as the collection of all variables not including ξ :

Algorithm 1 DGP-ETAS model for two GP layers

```

Initialize all variables from priors
for  $i = 1 : N$  do
   $w_1 \leftarrow \text{ESS}(p(w_1|\mathcal{C}, \Theta_{w_1}))$ 
   $(\mu_1, l_1, \sigma_1) \leftarrow \text{HMC}(p(\mu_1, l_1, \sigma_1|\mathcal{C}, \Theta_{\mu_1, l_1, \sigma_1}), l_s, n_s)$ 
   $w_2 \leftarrow \text{ESS}(p(w_2|\mathcal{C}, \Theta_{w_2}))$ 
   $(\mu_2, \sigma_2) \leftarrow \text{HMC}(p(\mu_2, \sigma_2|\mathcal{C}, \Theta_{\mu_2, \sigma_2}), l_s, n_s)$ 
   $B \leftarrow p(B|\mathcal{C}, \Theta_B)$ 
   $(K, \alpha, c, \tilde{p}) \leftarrow \text{HMC}(p(K, \alpha, c, \tilde{p}|\mathcal{C}, \Theta_{K, \alpha, c, \tilde{p}}), l_s, n_s)$ 
end for

```

Table 1. Example time to run each MCMC simulation (six chains of 100 000 members each) to the nearest second using eight threads of an Apple M1 Max CPU, with Julia 1.8.2. Note that only six threads are actively used for MCMC simulation, with the other two made available for redundancy and coordination, and that timings contain considerable variability (e.g. at least $\sim \pm 10$ per cent) due to computer resource allocation.

Scenario	Number of events	Constant rate	One layer	Two layer
Synthetic constant	330	513	1004	3703
Synthetic Gaussian	476	1090	3475	5234
Synthetic square	889	6074	7338	3515
Cahuilla swarm	461	1104	3055	5010
Ridgecrest sequence	920	6609	5792	7879

l_s and n_s are the hyperparameters that control the HMC integrator step length and number of integrator steps per sample, which we set to 0.01 and 10, respectively—empirically, these values result in good MCMC sampling efficiency for all examples.

6 EXAMPLES

As a first illustrative example, we investigate the inversion of synthetic earthquake catalogues, generated using the ETAS model simulated with Ogata's thinning algorithm (Ogata 1981). Three catalogues were generated, on a target interval of $T = 1000$ d length: a catalogue with constant background rate

$$\mu_{\text{const}}(t) = 0.2 \text{ d}^{-1}; \quad (16)$$

a catalogue with a Gaussian background rate

$$\mu_{\text{gauss}}(t) = \left(4.8 * \exp\left(-\frac{(t - T/2)^2}{2 * (\frac{T}{8})^2}\right) + 0.2 \right) / 5 \text{ d}^{-1}; \quad (17)$$

and a catalogue with a square wave background rate

$$\mu_{\text{square}}(t) = \begin{cases} 5 \text{ d}^{-1} & \frac{T}{4} < t < \frac{3T}{4} \\ 0.04 \text{ d}^{-1} & \text{otherwise} \end{cases}. \quad (18)$$

All catalogues were generated with an implicit cutoff magnitude $M_0 = 0$, $K_{\text{true}} = 0.2$, $\alpha_{\text{true}} = 1.25$, $p_{\text{true}} = 1.2$ and $c_{\text{true}} = 0.05$. The catalogue magnitudes follow the Gutenberg–Richter law with $b = 1.0$. In these synthetic experiments, as well as the real data experiments that follow, we generate $N = 100\,000$ samples for six independent MCMC chains, which we compare to ensure that the chains are individually well mixed. The time taken for the MCMC sampling to complete is detailed in Table 1.

Fig. 2 shows the output of the synthetic inversions for these three test catalogues, using the basic ETAS model, the single-layer GP model, and the two-layer DGP model. In this test, we are primarily interested in showing that the GP and DGP models do not overfit the constant model, and to observe their behaviour for variable background rate models. In the case of the constant model, we can see that the GP and DGP models do not seem to overfit compared to the ground truth background model. For the Gaussian background model, both the GP and DGP capture well the true model, as is expected given that the Gaussian perturbation can be well described with a single characteristic timescale for the GP.

For the square perturbation, the DGP captures the rapid onset of change in background behaviour at the edges of the square perturbation, indicating that it is learning the heterogeneity in the timescale of variations. Compared to the single-layer GP model, the edge is significantly sharper, indicating the improved performance of the two-layer DGP model in this case. Fig. 3 zooms into the case of the square wave perturbation with the DGP model, and shows the output posterior timescale l_2 ; we see that the characteristic timescale is shortened at the edges of the perturbation. The square example also best highlights the impact of model specification on the time taken for MCMC sampling

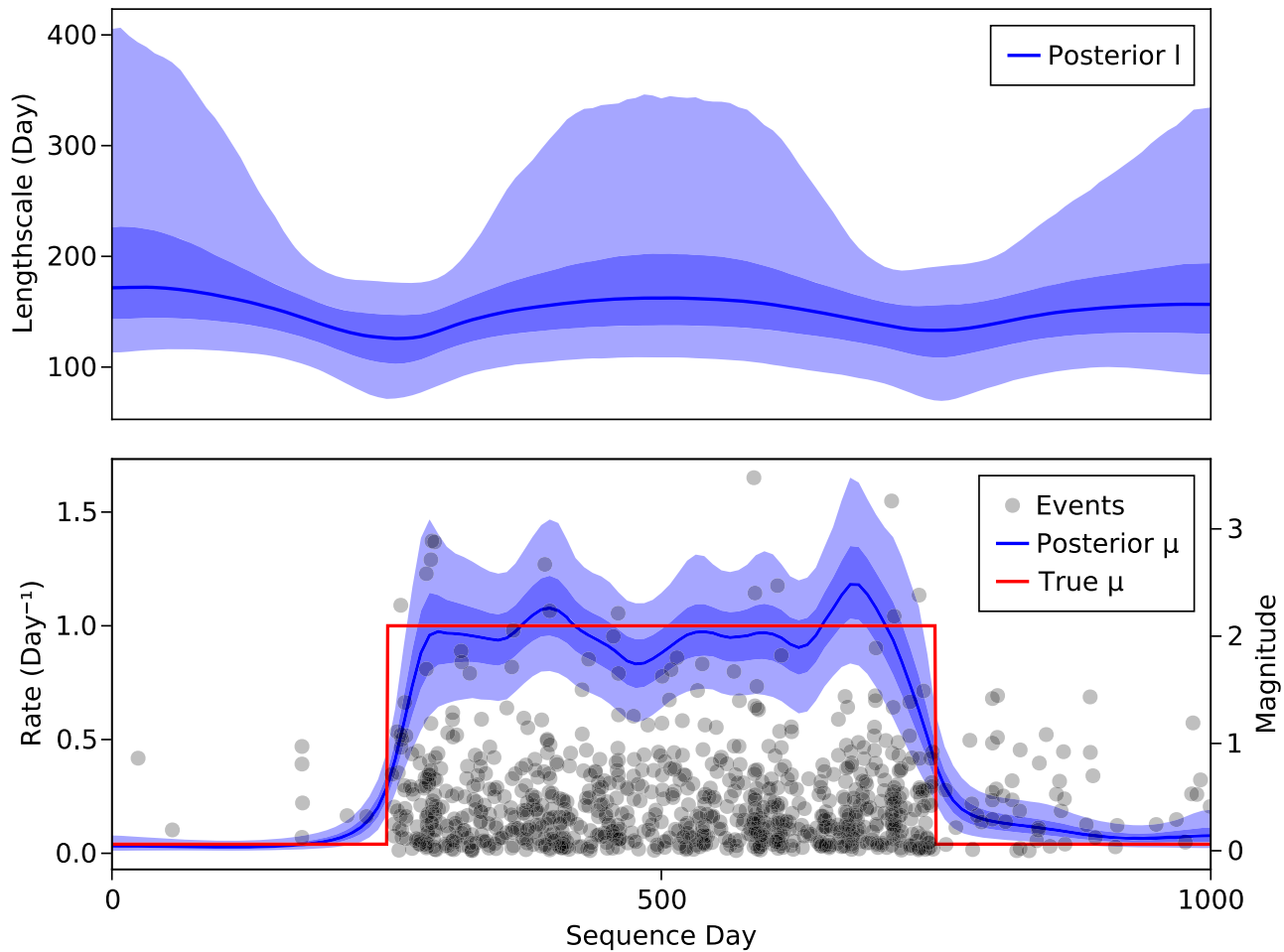


Figure 3. Results of the inversion for background rate of a synthetic catalogue using a square wave perturbation as the input. Catalogue events are shown as grey circles, the ground truth background rate as the red line, and the 90 percent credible interval, 50 percent credible interval and median of the posterior are shown in increasingly dark shades of blue.

to complete. If the model is mis-specified, then the posterior is typically difficult to traverse, leading to poor sampling performance; thus, the constant-rate model takes a similar time to sample as the one-layer model, despite the one-layer model having many more parameters, as it cannot well capture the true behaviour of the catalogue. The two-layer DGP model actually takes less time to compute than the single-layer GP model for this case, as it has an easier time capturing the rapidly changing behaviour at the edges of the square perturbation.

The evolution of MCMC parameters with iteration number for all test cases, also known as the MCMC chain plots, are found in the Supporting Information.

Having illustrated the properties of the DGP-ETAS model on synthetic data, we now turn to two examples for real data. First, we estimate a DGP-ETAS model for the background rate evolution of the Ridgecrest, CA M_w 7.0 2019 July 5 foreshock/aftershock sequence. High-resolution seismic catalogues show that this sequence qualitatively obeys a classical aftershock model (Shelly 2020), with rates of seismicity by eye appearing to follow two superimposed Omori–Utsu decay curves corresponding to triggering by the largest (M_w 6.4) foreshock and M_w 7.1 main shock. We therefore expect that the background rate estimate should be roughly constant, and that there be no external forcing to the Ridgecrest sequence. The catalogue of 920 events consists of Southern California Seismic Network (SCSN) located events with $M_L \geq 3$ with the time range of UTC 2019-07-01T00:00:00 to 2019-07-15T00:00:00 and between $(35.4699^\circ, 36.0361^\circ)$ latitude and $(-117.9435^\circ, -117.2132^\circ)$ longitude. To obtain an estimate for reference ETAS parameters, we used events in the same location between UTC 2000-01-01T00:00:00 and 2019-07-01T00:00:00 to run the classical MLE ETAS routine in SAPP (Statistical Analysis of Point Processes with R, Ogata 2006). Fig. 4 shows the catalogue and estimated background rate. We also show the smoothed total rate of earthquakes, defined similarly to Marsan *et al.* (2013) by looking at each earthquake, and finding the time elapsed for a window of 20 events either side of the central earthquake (i.e. 41 events total). As was hypothesized, we see no significant evidence of substantial background rate changes during this portion of the sequence, noting that the estimated daily background rate remains far below the actual rate of earthquake production even to the 90 per cent credible interval. The smooth increase after the two largest events may be attributed to the triggering effect of events that fall below the cut-off magnitude of the catalogue (and hence, implicitly, the minimum magnitude produced by the formulation of ETAS used in this study), but which are still large enough to trigger future earthquakes (Sornette & Werner 2005a, b).

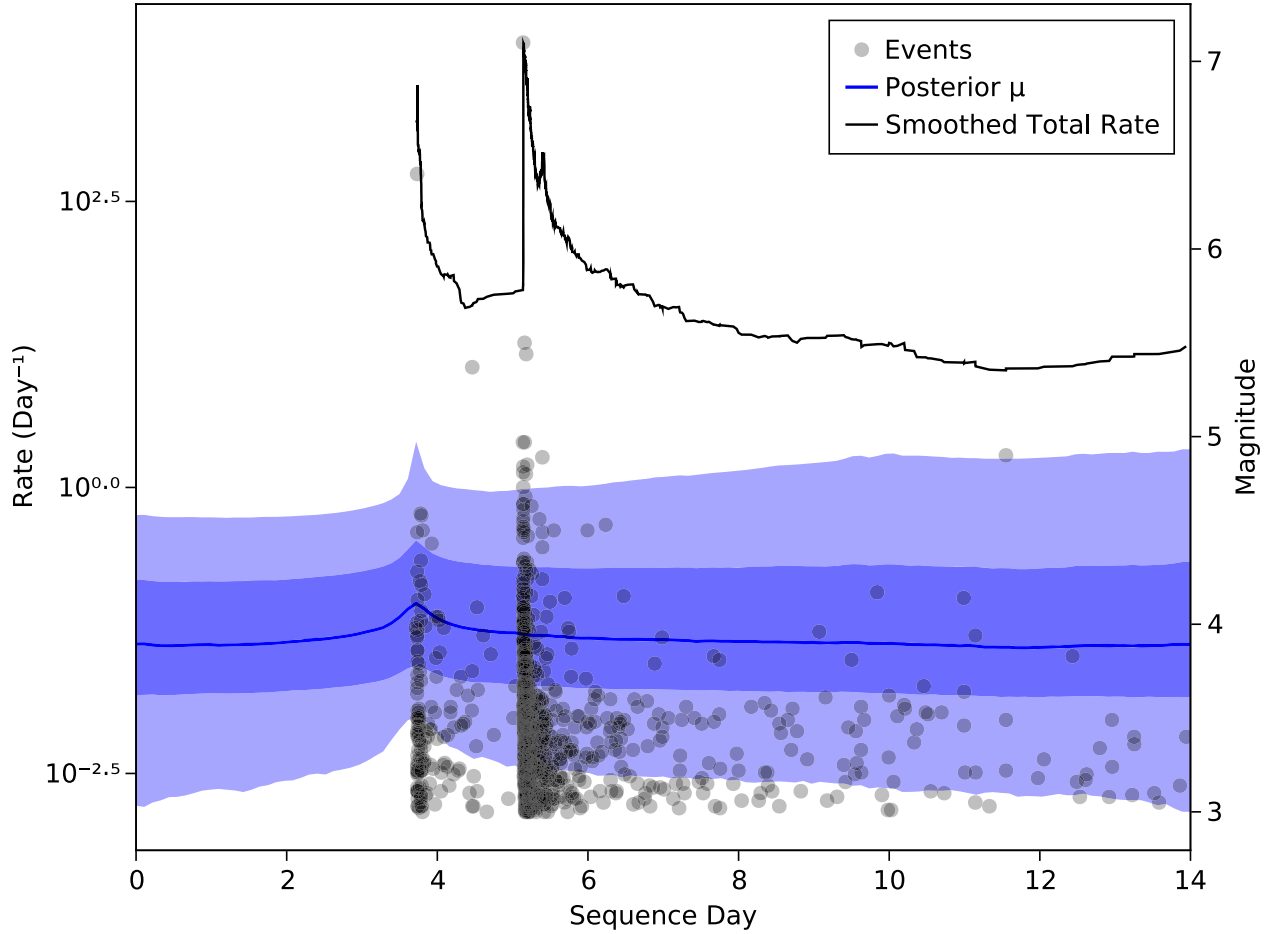


Figure 4. Results of the inversion for background rate of the Ridgecrest, CA M_w 7.0 2019 July 5 earthquake catalogue with a lower magnitude cut-off of 3.0. The results are shown for UTC 2019-07-01T00:00:00 to 2019-07-15T00:00:00. Catalogue events are shown as grey circles, and the 90 per cent credible interval, 50 per cent credible interval and median of the posterior are shown in increasingly dark shades of blue. The total earthquake rate, smoothed over a 41 event window, is shown in black. Due to the large difference between the total and background rates, the rates are plotted in log-space. As expected for an aftershock-driven event sequence, there is no large change in the expectation values of the background rate coincident with the two large events. The small increase observed may account for the triggering effect of earthquakes below the magnitude cut-off.

We then investigate a sequence with a totally different character—the Cahuilla, CA 2016–2019 earthquake swarm. This swarm qualitatively appears relatively unclustered prior to an M_w 4.4 earthquake late in the sequence, and instead has been hypothesized to be driven by the evolution of pore fluid pressure along the fault due to fluid injection (Ross *et al.* 2020). The catalogue of 461 events consists of SCSN events with $M_L \geq 1.71$ with time range UTC 2016-01-01T00:00:00 to 2019-12-31T00:00:00 and between $(33.42043^\circ, 33.58032^\circ)$ latitude and $(-116.84723^\circ, -116.68990^\circ)$ longitude. The reference background rate estimate was obtained in this case by assuming Poissonian earthquake statistics for the preceding time range of UTC 2000-01-01T00:00:00 to 2016-01-01T00:00:00. Fig. 5 shows the output. We note that the background rate estimate accelerates sharply around sequence day 500, and then slowly decays, before abruptly accelerating again after the largest M_w 4.4 earthquake before dying quickly away. The secondary peak is significant, and indicates increased seismicity after the M_w 4.4 event is not purely aftershock driven, although we note that the impact of aftershocks on the total rate is significant as indicated by the discrepancy between the smoothed total rate of earthquakes and the estimated posterior of μ for this second peak. Note that the smoothed total rate and the posterior μ are very similar for the first peak, indicating that the sequence is primarily driven aseismically prior to the M_w 4.4 event. This timeline matches well the sequence of events outlined by Ross *et al.* (2020), in which the rate of earthquakes increases geometrically with spreading fluid intrusion until reaching a physical barrier around day 600 at which point the swarm is geometrically constrained. Coincident with the M_w 4.4 event, the swarm circumvents this barrier, increasing the seismicity rate once more.

7 DISCUSSION AND CONCLUSIONS

We have illustrated the proposed DGP-ETAS model using both synthetic and real data sets, showing that it can capture the heterogeneous timescale structure in the background rate of earthquake sequences. It is pertinent to discuss the DGP-ETAS model in the context of existing methodology for investigating time variable background rates of seismicity. ETAS models have most commonly been solved using MLE, in which the negative log-likelihood is minimized using numerical optimization. MLE-based methods can incorporate time-varying background

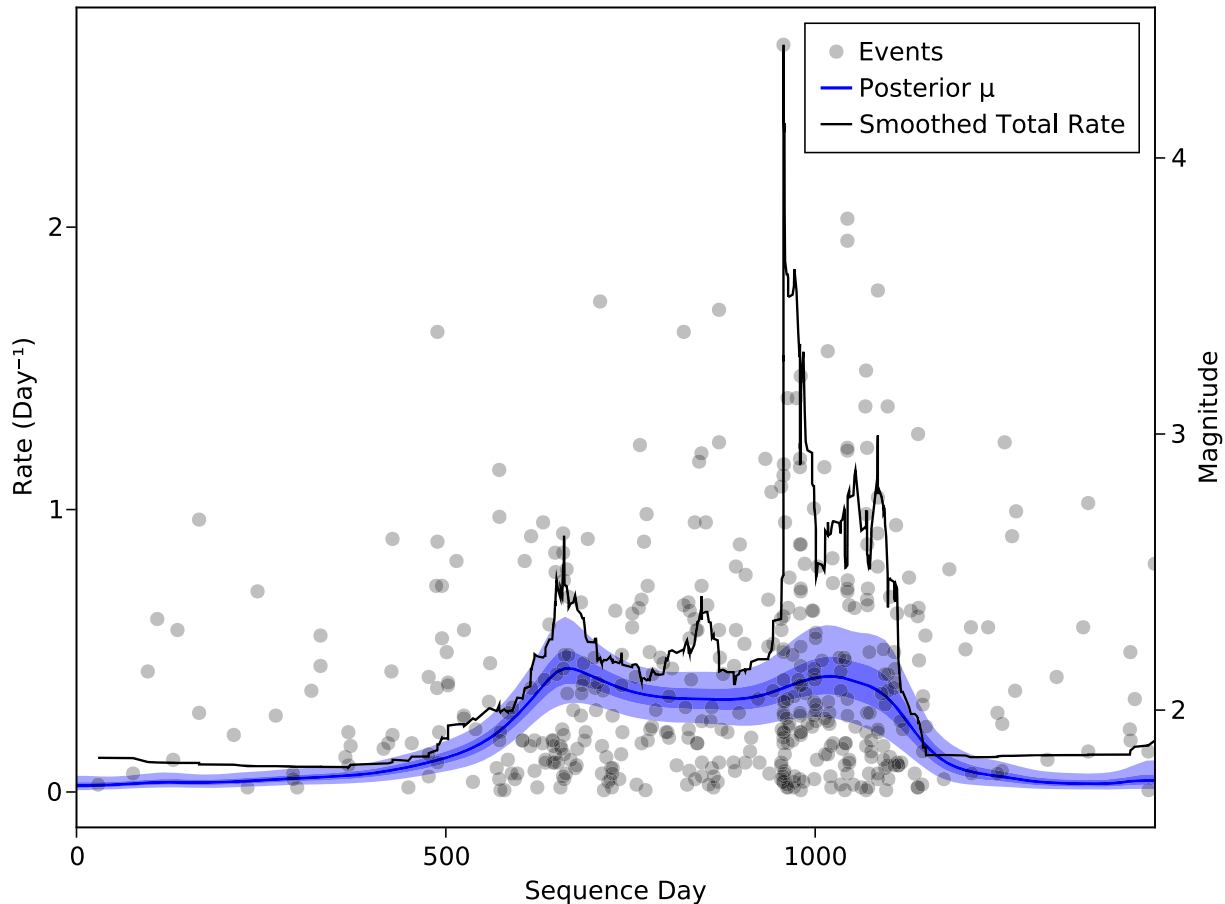


Figure 5. Results of the inversion for the background rate of the Cahuilla, CA earthquake swarm with a lower magnitude cut-off of 1.71. The results are shown for UTC 2016-01-01T00:00:00 to 2019-12-31T00:00:00. Catalogue events are shown as grey circles, and the 90 per cent credible interval, 50 per cent credible interval and median of the posterior are shown in increasingly dark shades of blue. The total earthquake rate, smoothed over a 41 event window, is shown in black. There is a significant change in the background rate associated with this swarm, including two peaks of activity, one of which appears to be associated with a subcluster of activity just before sequence day 1000.

rates $\mu(t)$ that are parametrized by functions such as piecewise constant, piecewise linear or splines. The complexity of these parametric forms can be controlled using information criteria methods and regularization (Kumazawa & Ogata 2014; Kumazawa *et al.* 2016). Alternatively, Marsan *et al.* (2013) suggest a semi-parametric approach, wherein a catalogue is alternatively fit with MLE ETAS given a prescribed forcing rate, and then forcing rate is updated and smoothed to best fit the catalogue with those ETAS parameters. The result is a background-rate model without a specific functional form (although the level of smoothing strongly impacts the results). MLE methods have the advantage of fast computation, although the poor numerical scaling of traditional ETAS formulations limits the catalogue size, and it is challenging to incorporate the branching variable formulation inside a numerical optimization framework. The MLE is a single point estimate, and does not provide robust uncertainty estimates for the parameters, including the background rate. Semi-parametric methods such as that of Marsan *et al.* (2013) still require multiple iterations for convergence. Additionally, the MLE cannot explicitly include *a priori* information about ETAS parameters, although a Bayesian *maximum a posteriori* (MAP) estimate can be similarly formulated to the MLE and solved to give a point estimate with prior information. A Bayesian formulation of ETAS, in contrast, gives access to uncertainty estimates for parameters, a robust framework for incorporating *a priori* knowledge, and can easily handle the branching variable formulation to accelerate computation of the ETAS log-likelihood. A GP-based Bayesian formulation (e.g. as proposed here and in Molkenthin *et al.* 2022) is semi-parametric and avoids making strong, explicit assumptions as to the shape of aseismic forcing, a feature that is useful for interpretation. It is interesting to observe the affinity of GP-based methods to the smoothing approach of Marsan *et al.* (2013), in which the smoothing window can be likened to a top-hat-shaped kernel. These benefits have to be weighed against the general cost of MCMC sampling, which is substantial but required to fully characterize the posterior. Additionally, from an end-use standpoint uncertainty bounds on the background rate are less critical than those on ETAS parameters in a forecasting setting, where accurate hazard probabilities are essential. The DGP-ETAS model proposed here combines some of the advantages of the semi-parametric formulations of Marsan *et al.* (2013) and Molkenthin *et al.* (2022), with the additional benefit of providing the characteristic rate-of-change of the aseismic forcing rate. The DGP framework amounts to a form of Bayesian kernel learning for GPs, and to some extent avoids the need for explicit construction of heterogeneity in parametrizing the GP kernel, which would otherwise rely on substantial expert knowledge. We have also shown how the use of the SPDE approximation for

GPs reduces the computational complexity for the problem and naturally allows for the inclusion of layered GP structure. Thus, the DGP framework based on SPDE approximations answers two of the key questions in Molkenhuth *et al.* (2022) that hindered the applicability of GPs as a basis for semi-parametric modelling of earthquake sequences. The next immediate extension of the framework presented in this paper is to include spatial variability of the background rate—the examples presented in this paper are purely temporally varying and for the real data examples consist of very small geographic regions. There is no inherent change in the theory of using DGPs in higher dimensions and the results presented here would transfer directly even in the 3 + 1-D case—however, the efficiency of the SPDE approximation declines relative to classical GPs as dimensionality increases, as PDE solutions in higher dimensions necessarily must contend with geometrically increasing volumes whereas GPs scale purely with the data. A possible way to circumvent this issue could be to use a grid-refined finite-element approach for solving the SPDE (Rue *et al.* 2009, e.g.), which may scale better to high dimensions than our simple finite-difference approach. This is especially true if the seismicity is highly clustered so that empty parts of the space–time domain can be meshed with very large elements. A method for fitting ETAS models with background rates defined on finite-elements and a Laplace approximation to the posterior is investigated in Naylor *et al.* (2023). Alternatively, other methods of constructing DGPs which do not rely on the covariance operator approach advocated here may scale better (Paciorek & Schervish 2003, e.g.), although they lose the attractive interpretability of the intermediary timescales in the DGP presented here. For particularly large catalogues, it is likely that further approximation of the GP background rate and ETAS parameters via variational methods may be required, as ETAS remains in general an expensive model to compute—a problem compounded by exact sampling methods such as MCMC. Ray & Myer (2019) and Blatter *et al.* (2021) have investigated an approximate GP-like hierarchical method for magnetotelluric tomography, which utilizes node points throughout the inversion domain that are added parsimoniously using trans-dimensional methods. Whilst their scheme does not account for the full hierarchical posterior of a deep-GP model, it does illustrate that the class of models discussed here may be useful for tomography problems as well as point processes.

In summary, the DGP-ETAS model allows detailed quantitative statistical analysis of earthquake sequences under very general assumptions for the background rate, and we have explored and validated its applicability for a range of interesting cases. The DGP framework is highly flexible, and may be applied to other geophysical problems in which we expect heterogeneity in the characteristic length-scales and timescales for variation of a parameter of interest.

DATA AND CODE AVAILABILITY

The AseismicGP.jl package, used to run all inversions in this paper, may be installed for Julia 1.6+ by first adding <https://github.com/jbmuir/JBMuirJuliaRegistry.git> to the list of registries and then adding AseismicGP using the package manager. All figures in the paper (other than the schematic Fig. 1) may be reproduced by running the demo code at <https://github.com/jbmuir/AseismicGPDemo>, which automatically installs the required registry. Catalog data for the Ridgecrest and Cahuilla sequences was downloaded from the Southern California Earthquake Data Center (SCEDC, 2013).

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REFERENCES

- Bayliss, K., Naylor, M., Illian, J. & Main, I.G., 2020. Data-driven optimization of seismicity models using diverse data sets: generation, evaluation, and ranking using inlabru, *J. geophys. Res.: Solid Earth*, **125**(11), e2020JB020226.
- Betancourt, M., 2017. A conceptual introduction to hamiltonian monte carlo, arXiv:1701.02434 [stat].
- Blatter, D., Ray, A. & Key, K., 2021. Two-dimensional bayesian inversion of magnetotelluric data using trans-dimensional gaussian processes, *Geophys. J. Int.*, **226**(1), 548–563.
- Damianou, A. & Lawrence, N.D., 2013. Deep gaussian processes, in *Proceedings of the Sixteenth International Conference on Artificial Intelligence and Statistics*, pp. 207–215, PMLR.
- Dunlop, M.M., Girolami, M.A., Stuart, A.M. & Teckentrup, A.L., 2018. How deep are deep gaussian processes?, *J. Mach. Learn. Res.*, **19**(1), 2100–2145.
- Gelfand, A.E. & Schliep, E.M., 2016. Spatial statistics and gaussian processes: a beautiful marriage, *Spatial Stat.*, **18**, 86–104.
- Hainzl, S., Zakharova, O. & Marsan, D., 2013. Impact of aseismic transients on the estimation of aftershock productivity parameters, *Bull. seism. Soc. Am.*, **103**(3), 1723–1732.
- Kumazawa, T. & Ogata, Y., 2014. Nonstationary ETAS models for nonstandard earthquakes, *Ann. Appl. Stat.*, **8**(3), 1825–1852.
- Kumazawa, T., Ogata, Y., Kimura, K., Maeda, K. & Kobayashi, A., 2016. Background rates of swarm earthquakes that are synchronized with volumetric strain changes, *Earth planet. Sci. Lett.*, **442**, 51–60.
- Lindgren, F., Rue, H. & Lindström, J., 2011. An explicit link between Gaussian fields and Gaussian Markov random fields: the stochastic partial differential equation approach: link between Gaussian Fields and Gaussian Markov Random Fields, *J. R. Stat. Soc.: Ser. B (Stat. Methodol.)*, **73**(4), 423–498.
- Marsan, D. & Enescu, B., 2012. Modeling the foreshock sequence prior to the 2011, MW9.0 Tohoku, Japan, earthquake, *J. geophys. Res.: Solid Earth*, **117**(B06316).
- Marsan, D., Prono, E. & Helmstetter, A., 2013. Monitoring aseismic forcing in fault zones using earthquake time series, *Bull. seism. Soc. Am.*, **103**(1), 169–179.

- Molkenthin, C., Donner, C., Reich, S., Zöller, G., Hainzl, S., Holschneider, M. & Oppen, M., 2022. GP-ETAS: Semiparametric Bayesian inference for the spatio-temporal epidemic type aftershock sequence model, *Stat. Comput.*, **32**(29), 1–25.
- Murray, I., Adams, R. & MacKay, D., 2010. Elliptical slice sampling, in *Proceedings of the Thirteenth International Conference on Artificial Intelligence and Statistics*, pp. 541–548, JMLR Workshop and Conference Proceedings.
- Naylor, M., Serafini, F., Lindgren, F. & Main, I., 2023. Bayesian modelling of the temporal evolution of seismicity using the ETAS.inlabru R-package. *Front. Appl. Math. Stat.*, **9**, doi:10.3389/fams.2023.1126759.
- Neal, R.M., 2011. MCMC using Hamiltonian dynamics, *Handbook of Markov Chain Monte Carlo*, Vol. 2(11). Chapman and Hall/CRC
- Ogata, Y., 1981. On Lewis' simulation method for point processes, *IEEE Trans. Inform. Theory*, **27**(1), 23–31.
- Ogata, Y., 1988. Statistical models for earthquake occurrences and residual analysis for point processes, *J. Am. Stat. Assoc.*, **83**(401), 9–27.
- Ogata, Y., 1998. Space-time point-process models for earthquake occurrences, *Ann. Inst. Stat. Math.*, **50**(2), 379–402.
- Ogata, Y., 2006. *Statistical Analysis of Seismicity: Updated Version (SA-Seis2006)*, Vol. 33 of Computer Science Monographs, The Institute of Statistical Mathematics, Tokyo.
- Paciorek, C. & Schervish, M., 2003. Nonstationary covariance functions for gaussian process regression, in *Advances in Neural Information Processing Systems*, Vol. 16, MIT Press.
- Rasmussen, C.E. & Williams, C.K.I., 2006. *Gaussian Processes for Machine Learning*, Adaptive Computation and Machine Learning, MIT Press, Cambridge, MA.
- Ray, A. & Myer, D., 2019. Bayesian geophysical inversion with trans-dimensional Gaussian process machine learning, *Geophys. J. Int.*, **217**(3), 1706–1726.
- Roininen, L., Girolami, M., Lasanen, S. & Markkanen, M., 2016. Hyperpriors for Matérn fields with applications in Bayesian inversion, arXiv:1612.02989 [math, stat].
- Ross, G.J., 2021. Bayesian estimation of the ETAS model for earthquake occurrences, *Bull. seism. Soc. Am.*, **111**(3), 1473–1480.
- Ross, Z.E., Ben-Zion, Y. & Zaliapin, I., 2022. Geometrical properties of seismicity in California, *Geophys. J. Int.*, **231**(1), 493–504.
- Ross, Z.E., Cochran, E.S., Trugman, D.T. & Smith, J.D., 2020. 3D fault architecture controls the dynamism of earthquake swarms, *Science*, **368**(6497), 1357–1361.
- Ross, Z.E., Idini, B., Jia, Z., Stephenson, O.L., Zhong, M., Wang, X., Zhan, Z., Simons, M., *et al.*, 2019. Hierarchical interlocked orthogonal faulting in the 2019 Ridgecrest earthquake sequence, *Science*, **366**(6463), 346–351.
- Rue, H., Martino, S. & Chopin, N., 2009. Approximate Bayesian inference for latent Gaussian models by using integrated nested Laplace approximations, *J. R. Stat. Soc.: Ser. B (Stat. Methodol.)*, **71**(2), 319–392.
- Shelly, D.R., 2020. A high-resolution seismic catalog for the initial 2019 Ridgecrest earthquake sequence, *Seismol. Res. Lett.*, **91**(4), 1971–1978.
- Shelly, D.R., Ellsworth, W.L. & Hill, D.P., 2016. Fluid-faulting evolution in high definition: Connecting fault structure and frequency-magnitude variations during the 2014 Long Valley Caldera, California, earthquake swarm, *J. geophys. Res.: Solid Earth*, **121**(3), 1776–1795.
- Sornette, D. & Werner, M.J., 2005a. Apparent clustering and apparent background earthquakes biased by undetected seismicity, *J. geophys. Res.: Solid Earth*, **110**(B09303).
- Sornette, D. & Werner, M.J., 2005b. Constraints on the size of the smallest triggering earthquake from the epidemic-type aftershock sequence model, Bath's law, and observed aftershock sequences, *J. geophys. Res.: Solid Earth*, **110**(B08304).
- Southern California Earthquake Data Center (SCEDC), 2013, doi:10.7909/C3WD3xH.
- Utsu, T., 1961. A statistical study on the occurrence of aftershocks, *Geophys. Mag.*, **30**, 521–605.
- Utsu, T., 1971. Aftershocks and earthquake statistics (2): further investigation of aftershocks and other earthquake sequences based on a new classification of earthquake sequences, *J. Fac. Sci., Hokkaido Univ. Ser. 7, Geophys.*, **3**(4), 197–266.
- Veen, A. & Schoenberg, F.P., 2008. Estimation of space–time branching process models in seismology using an EM-type algorithm, *J. Am. Stat. Assoc.*, **103**(482), 614–624.
- Wei, S., *et al.*, 2011. Superficial simplicity of the 2010 El Mayor–Cucapah earthquake of Baja California in Mexico, *Nat. Geosci.*, **4**(9), 615–618.

SUPPORTING INFORMATION

Supplementary data are available at [GJI](https://doi.org/10.1017/gji.2023.1126759) online.

Figure S1 MCMC chains for synthetic constant model, inverted using a constant background rate.

Figure S2 MCMC chains for synthetic constant model, inverted using a one-layer background rate.

Figure S3 MCMC chains for synthetic constant model, inverted using a two-layer background rate.

Figure S4 MCMC chains for synthetic Gaussian model, inverted using a constant background rate.

Figure S5 MCMC chains for synthetic Gaussian model, inverted using a one-layer background rate.

Figure S6 MCMC chains for synthetic Gaussian model, inverted using a two-layer background rate.

Figure S7 MCMC chains for synthetic square model, inverted using a constant background rate.

Figure S8 MCMC chains for synthetic square model, inverted using a one-layer background rate.

Figure S9 MCMC chains for synthetic square model, inverted using a two-layer background rate.

Figure S10 One-layer inversion for the Ridgecrest sequence.

Figure S11 One-layer inversion for the Cahuilla swarm.

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